SPMM Smart Revise

Human Development

Paper A Syllabic content 2.1 to 2.18

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1. Conceptualizing development

In psychology, **maturity** refers to the ability to respond to the environment in an appropriate manner, usually with a learnt response. Maturity refers to the production of expected behaviour and actions in a given situation in an age-appropriate manner. Psychosocial maturity is a key factor in the development of a sense of autonomy and includes processes involving cognitive, emotional, social and moral development.

Nature vs. Nurture: Development is influenced by both nature (genetic disposition) and nurture (environmental influences). But the proportional contribution of genetics and environment to the psychosocial development of personality, intelligence and sexuality is fiercely debated to date.

Studies estimate a heritability of 40-50% for human IQ. But several observations warrant consideration in this regard.

1. *Heritability of IQ varies with age*: While the heritability is around 30% in children, it increases to 80% among adults. This suggests that either the genetic determinants of intelligence are age-specific or that the non-genetic maturational factors are much more influential in shaping human intelligence at younger ages.
2. *Heritability of IQ is affected by demographics*: Scarr et al. observed greater genetic effects on intelligence in middle-class white groups than in lower-class African American groups, suggesting that among lower socioeconomic groups, nongenetic influences operate on development. Irving Gottesman, the proponent of endophenotype concept, famously stated that *genes are weaker than poverty*.
3. *Genetic influences are likely to be variable*: For a construct such as intelligence, which is made of several sub-constructs, it is likely that no single gene or genetic complex will be sufficient to account for the variations. Furthermore, at various stages of development, the genetic factors operating to influence intelligence could vary.
4. *Effect of the shared environment*: genetic influences are often inferred from the observation that closer biological relatives (e.g. identical twins) are more similar in their intelligence than less closely related pairs (non-twin siblings). But, in fact, several observations suggest that there is indeed a greater similarity between pairs of family members than would be predicted on the basis of their biological relationship, indicating the effect of shared environment on intelligence (Deary et al., 2010).

Models and theories

Developmental theories aim to explain how children grow and learn. Of these stage, theories refer to theories that consider development as a process that occurs over a 12- to the l5-year period in chunks of time called **stages**. Within each stage, a specific set of functioning and behaviour can be observed. Certain **maturational tasks** (motor, cognitive and perceptual) are also achieved in each stage, heralding transition to next stage of development. Piaget's theory is a prototype stage theory focused on **epistemology** (the study of the development of knowledge or intelligence). According to Piaget, development

1. Occurs in hierarchical stages
2. Each stage is qualitatively different (not just quantitatively)
3. The stages consist of invariant functions and all children undergo these sequentially

Other stage theorists include Gesell, Freud and Erikson.

Maturational tasks

Age	Motor	Language	Sensory	Social development
4-6 weeks				Smiles at the parent (social smile - 6 weeks); can recognise mum's face apart; shows preference to human faces.
6-8 weeks		Cooing		
3 months	Can hold head up. grasp reflex disappears	Babbling	Localises sound source	Squeals with pleasure appropriately. Discriminates smile
5 months	Reaches out; oral exploration	Spontaneous babbling and sound experiments		
6 months	Hand to hand transfer rolling over Palmar grasp	Double syllable sounds such as 'dada.'	Localises sound 45cm lateral to either ear	
9-10 months	Cruises around and crawls. Sits unsupported. Picks up objects with pincer grasp	Babbles tunefully	Looks for toys dropped; Peek a boo	Stranger anxiety followed by object permanence

			game	
1 year	Stands alone momentarily	One or two words		Separation anxiety
18 months	Walks alone. Holds rails and climbs, can jump with both feet. Can build a tower of 3 or 4 cubes and throw a ball (1 X 3). Can use a spoon.	Many intelligible words – up to 40 in some. Uses holophrases.		Shows rapprochement (hugs when coming back).
2 years	Able to run. Builds tower of 6 cubes (2 X 3)	Makes sentences – telegraphic initially.		Parallel play. Dry by day
3 years	Goes upstairs 1 foot per step and downstairs 2 feet per step. Copies circle, imitates cross and draws the man on request. Builds tower of 9 cubes (3 X 3)	Speaks in sentences		Cooperative play. Imaginary companions
4 years	Can skip; copies a cross			Toilet trained mostly
5 years	Can hop; copies a triangle.	Fluent speech with grammar use; uses function words		Dresses and undresses alone
6 years	Copies a diamond. Can count number of fingers	Nearly adult-like speech		

A brief outline of various developmental theories is presented in the table below. These will be considered in detail in later sections of this chapter.

Theories	Key Concepts
Temperament theory (Thomas & Chess, Kagan)	Temperaments are inherent biologically based traits varying from difficult to easy (or inhibited to uninhibited). Temperament elicits environmental response that perpetuates a pattern of behaviour
Organismic stage theory (Piaget)	Development occurs in stages with transition occurring as a result of interaction of the child with its environment (child as a scientist)
Attachment theory (Bowlby)	Innate tendency to seek relationships influence patterns of behaviours in later life.
Social learning theory (Bandura)	Observational learning in childhood influences later behaviour
Psychosexual stage theory (Freud)	Stage-specific behaviours are driven by inner conflicts and resulting anxiety signals. Successful resolution of conflicts aid in progressive maturation.
Psychosocial stage theory (Erikson)	Psychosocial developmental stages are characterized by conflicts, but the successful resolution is not mandatory for further development.
Collaborative learning theory (Vygotsky)	Development is not entirely private; child acts as an apprentice in social surroundings rather than a scientist. Parents and teachers carry out the role of **scaffolding** to introduce familiarity for the child to develop its own expertise (**collaborative learning**). **Zone of proximal development** refers to functions that are not yet fully achieved but are in the process 'pipeline' whose development is aided by scaffolding.
Maturational Growth theory (Gesell)	Maturation of the nervous system as the principal driver of the various aspects of human behaviour

Freud's psychosexual stages: Gradual, the sequential emergence of genital sexuality from infantile sexuality is noted in Freud's model. The stages discussed here reflect both biological and psychological maturation.

Freud's Psychosexual Stage	Characteristics
ORAL (0 to 1 ½ years)	Drive discharge is via sucking; oral erotogenic zone. oral erotism (sucking, licking, etc.) in early stages; oral sadism (biting, chewing) in later stages. The ego develops at this stage.
ANAL (1 ½ to 3 years)	Anal erotogenic zone; drive discharge via sphincter behaviour. Anal erotism refers to the sexual pleasure in anal functioning. Anal sadism refers to the aggressive wishes linked to fecal expulsion. Anal fixation is characterized by OCD like pattern – also ambivalence and sadomasochistic tendencies are associated.
PHALLIC/OEDIPAL (3 to 5 yrs)	Genitals become organs of interest; masturbation-like activity noted. Oedipus complex – wish to have a libidinal relationship with opposite sex parent (Electra complex in girls) with a desire to exclude the rival parent. This lead to a fear of retaliation from the rival parent in the form of castration anxiety in boys and loss of mother's love in girls. Electra complex in girls include penis envy, a wish to have penis is accompanied by blaming the mother for absence of penis; later this becomes a secret wish to displace mother as object of father's love and bear his baby. At the resolution of Oedipus and Electra complexes, identification with the aggressor i.e. dad for a boy and mum for a girl occurs; super-ego develops from introjection of parental values. Abraham divided this into early partial genital (true phallic phase) and later mature genital phase.
LATENCY (5 to puberty approx.11yrs)	Socialization, interest in peers seen. Sexual energy sublimated towards school work, hobbies and friends
GENITAL (puberty onwards)	Biological maturation occurs; genital sexuality is born.

Adversities and development

A **critical period** is a time point when an individual is acutely sensitive to the effects of external influences - both positive and negative. This is usually defined by biological and psychosocial events. This concept is related to the notion that there is a gradually decreasing plasticity in functioning across the life span. But this is challenged by some observations that suggest, for example, that maturational tasks such as an attachment can be formed even at later ages.

Early life is the period of most rapid brain development. Therefore, this period is a sensitive phase for both positive and adverse factors to influence human development. Severe neglect (e.g. in relation to institutional care) produces adverse consequences if it occurs in early rather than later childhood. Similarly, the effects of toxins (e.g. lead and alcohol are far more dramatic when the exposure occurs in utero or in early life. Apart from early life, adolescence is also another critical period in life. Major life transitions influencing development occur during adolescence.

On the basis of potential to cause enduring physiologic disruptions, 3 distinct types of stress responses are described in young children.

STRESS VULNERABILITY MODEL

Zubin & Spring (1977) proposed the stress vulnerability model. According to this model mental illness, schizophrenia especially, is a result of two hits. The first hit is the vulnerability or predisposition of an individual that may be biologically or psychosocially determined. The second hit is the stress factor, which may act as a trigger or precipitant. This could also be biological, psychological or social. Low vulnerable individuals will require high degree of stress to develop an illness while highly vulnerable may respond to hairline triggers.

In an interesting study of environment-gene interaction, Caspi et al (2003) noted that individuals with one or two copies of the short allele of the 5-HT T promoter polymorphism exhibited more depressive symptoms, diagnosable depression, and suicidality in relation to stressful life events than individuals homozygous for the long allele.

- **Positive stress response** – brief, mild response moderated by the availability of a caring and responsive adult. e.g. getting an immunization, anxiety associated with the first day at a nursery. When buffered adequately positive stress responses are growth-promoting opportunities.
- **Tolerable stress response** - associated with exposure to non-normative experiences with a greater magnitude of adversity. The e.g. death of a family member, a serious illness or injury. When buffered well the risk of physiologic harm and long-term consequences is greatly reduced.

- **Toxic stress response** - strong, frequent, or prolonged activation of the body's stress response in the absence of the buffering protection from supportive adults. e.g., child abuse or neglect, parental substance abuse, and maternal depression. Toxic stress disrupts the developing brain circuitry during sensitive developmental periods forming the precursors of later physical and mental illness.

Methodology for studying development

Children of different ages can be compared on a given behaviour in order to determine age-specific developmental features (**cross-sectional design**). But, in this case, inter-individual variations can distort the true picture. **Longitudinal designs** are often employed where the same children are studied twice or more over a prolonged period of time with respect to a developmental feature. This provides more reliable estimates of development though this can be time-consuming.

As in other scientific disciplines, in developmental psychology, the identification of a risk factor does not necessarily imply causation as the relationship could be mediated by a third variable (**mediator**), the putative risk factor may indeed be the outcome (**reverse causality**) or due to the presence of **multivariate relationships**. Such multivariate factors that partially account for the relationship between a risk factor and a disorder are termed mediators. The mediator could be a protective factor or a buffer that reduces the risk in an individual.

2. Attachment theory

Bowlby's views: According to Bowlby, attachment begins in infancy and lasts throughout a lifetime. A newborn baby immediately needs someone to take care of them. This person may be a parent, a sibling, or a nanny, but whoever it is, there will be a bond formed between them. Bowlby believed that this **primary caregiver** is the one that will most shape the child's personality and character. The primary caregiver is usually the mother (but need not always be), and strong bonds are formed within minutes of giving birth. It is important for the new parents and baby to be alone together right after the birth to establish a strong bond. If there are too many individuals in the room right after birth, the natural process of attachment can be disrupted and this can have long-term effects on the relationship between the child and parents (Klaus, Kennell, & Klaus, 1995).

The attachment formation needs **caregiver's presence** in early stages; no difference is made if motherly care is provided late after 30m especially and not in early stages. Attachment behaviour is more evident when distress is present.

According to Bowlby, the strong innate tendency to attach to one adult female is seen – this is called **monotropy**. This attachment is qualitatively different from later attachments made. But it is shown that multiple attachments are the rule rather than the exception. Around 18m, 87% infants have multiple attachments; 50% primarily attached to the mother, 18% to father and the rest to equally both. Attachment process itself is more important than who the attachment figure is.

Bowlby believed that attachment is **innate and adaptive**. We are all born with an inherited need to form attachments, and this is to help us survive. In his terms, the newborn infant is helpless and relies on its mother/caregiver for food, warmth, etc. and hence the attachment behaviour is essentially adaptive. Attachment behaviour peaks between 12-18 months but various phases are notable during development.

1. **Preattachment** phase (birth to 8 or 12 weeks), babies orient to their mothers,
2. **Indiscriminate** attachment (attachment in making - 8 to 12 weeks to 6 months): Allows strangers to handle, infants become attached to one or more persons in the environment
3. **Clear-cut attachment** (6 through 24 months): Preferential attachment, separation anxiety, object permanence, stranger anxiety. At the later part, weakened stranger anxiety; other attachment figures may also present.
4. After 25 months, the mother figure is seen as independent.

Harlow's experiments: These experiments established the importance of contact comfort as basic as the need for food in developing mother-infant bonding. Harlow separated rhesus monkeys

from their mothers during their first weeks of life. Harlow substituted a **surrogate mother** made from wire or cloth for the real mother. The infants preferred the cloth-covered surrogate mother, which provided contact comfort, to the wire covered surrogate, which provided no contact comfort. This preference was observed irrespective of feeding, i.e. the terry-cloth soft-surrogate mother was preferred even if it did not have a feeding nipple attached to it.

Ainsworth's experiments: Ainsworth constructed a **strange situation experiment** with 2 separation and 2 reunion episodes. An infant is observed in the presence and absence of its mother and a stranger in the vicinity in seven different combinations. According to the infant's behaviour it is classified as type A, B or C.

Strange Situation Experiment	
Situation 1	Both mother and infant enter the room
Situation 2	A stranger joins them
Situation 3	Mother leaves now; infant left with stranger
Situation 4	Mother returns; stranger leaves
Situation 5	Infant left alone; mother leaves now
Situation 6	Stranger comes back and tries to comfort the child
Situation 7	Mother comes back and comforts, stranger leaves.

This classification below correlates highly with 1. Responsiveness and sensitivity of the mothers to the needs of their children and 2. Total amount and quality of stimulation (holding) provided by the mothers.

- Type A: **Anxious avoidant**: 15%. Indifferent attitude to the mother is leaving the room or entering the room; keeps playing indifferent to mother's presence. Distress when alone, not when the mother is leaving. Stranger can comfort the child easily. Highly environment directed, low attachment behaviour. Greater in the West. Perpetrators of bullying mostly have this pattern.
- Type B: **Secure**: 70%. Plays independently when the mother is in the vicinity (secure base effect). Distress when the mother is leaving; seeks contact on the return of the mother and gets quickly comforted by the mother, not a stranger.
- Type C: **Anxious resistant**: 15%. Fussy and cries a lot and cannot use the mother as a secure base to explore around. Very high levels of distress are seen when the mother is leaving. But not comforted easily even on her return; appears ambivalent about her return. Active resistance to stranger's efforts to pacify. Highly caregiver directed low play behaviour. Greater in Japanese and Israeli families. Furthermore, this pattern is also common among victims of bullying.

- In some cases a fourth type D - **disorganised** type - is also seen. This is seen in maltreated or maternally deprived children. The child has an insecure, dazed look and acts as if it is frightened of the mother. This pattern may be a precursor to later personality difficulties or dissociative experiences. Mother may have an experience of being abused as a child.

Attachment style may differ with different caregivers; it is a function of the quality of caregiving and NOT the temperament of a child.

Main devised a semi-structured adult attachment interview with 15 items (**AAI**). This is based on the fact that infantile attachment pattern can be predicted reasonably accurately using discourse analysis of adults when recollecting their childhood. Accordingly 4 patterns are noted.

- Secure **autonomous**: Those who had secure attachment provide spontaneous and coherent answers with the ability to talk freely about negative experiences in childhood type B Ainsworth.
- **Dismissing** of experiences: Those who had an avoidant (insecure) pattern often minimise their experiences, do not elaborate on them and do not use colourful metaphors during the discourse– type A (avoidant)
- **Entangled**: Those who had insecure but ambivalent (enmeshed) attachment use multiple emotionally laden responses and ramble excessively, – type C resistant.
- Unresolved **disorganised**: Broken continuity and interrupted the logical flow of thoughts is seen in those who had insecure disorganised attachment pattern– type D.

The secure attachment appears to be a protective factor for the development of childhood disorders, and insecure attachment is best conceptualized as a risk factor for a number of childhood disorders. It has been demonstrated in various studies that insecure attachment during early childhood is associated with the development of behavioural problems especially *oppositional defiant disorder* at school age. Insecure attachment in combination with other vulnerability factors such as family dysfunction, difficult child temperament, and poor parental management can give rise to later childhood disorders

Spitz – anaclitic depression or **hospitalism**: When children are hospitalised for physical problems, a short period of separation from primary caregiver ensues; this loss of loved one is called anaclitic (object loss) depression. It is counterproductive to child's development. But recovery is good if the maternal deprivation is kept minimum i.e. less than 3 months. Rare if prolonged. Surrogate mothering helps the infant when having the anaclitic depression to some extent.

MAHLER'S STAGES
1. **Normal autism** (0 to 2 m): Child spends most time in sleep as if the intrauterine aloofness continues.
2. **Symbiosis** (2 to 5m): Inner and outer world studied via senses but perceives mother and self as one unit.
3. **Separation – individuation phase: (DPRO)** a) **Differentiation** sub-phase: (5 to 10m) slowly appreciates the difference between mother and self b) **Practicing** sub-phase: (10 to18m) A gradual increase in interest on the environment; practices exploration. c) **Rapprochement** sub-phase: (18 to 24m) Alternating drives to be autonomous and dependent; Able to explore alone but requires comfort and reassurance on return. d) **Object** constancy sub-phase: (2 to 5yrs) Understand that the mother will not be lost if temporarily away; hence able to function independently.

Margaret Mahler described the development of a sense of identity in young children, independent of their mothers. This is called **separation-individuation theory,** and the proposed stages are supposed to be universal in all children.

Rutter distinguished deprivation from privation.

- ✶ **Deprivation:** Attachment is formed but lost temporarily. If it is for a short time then protest – despair – detachment phases (similar to grief) are seen. This is more common in 8m to 3 yr. old. Boys show more deprivation features than girls. It is more noticeable if aggressive caregiving e.g. physical abuse was present before separation. In prolonged deprivation, separation anxiety sets in. Increased clingy behaviour, psychosomatic complaints, vacillation and aggression are seen in the child.
- ✶ **Privation** refers to the non-formation of attachment; this is very rare and can lead to what Rutter termed as '**affectionless psychopathy'** and developmental retardation. Attention seeking, lack of guilt, antisocial behaviour and indiscriminate attachment patterns are noted. This is reversible but only to some extent.

Ethology is the systematic biological study of animal behaviour. Greek ethos - custom or habit. It was coined by Heinroth. **Imprinting** is a special primitive form of learning wherein during the early period of development (called **critical** or **sensitive phase**) a young animal is highly sensitive to a certain stimulus that provokes a specific behaviour pattern. **Lorenz** described the **imprinting** in goslings where a moving object in the early period of development provokes the following behaviour of that moving object. This is useful as almost always mother is the first moving object for goslings and hence they learn to follow the mother; but when Lorenz disrupted

this by presenting himself as the moving object, the goslings imprinted by Lorenz followed him and refused to follow mother goose. Imprinting is particularly resistant to change.

Innate releasing mechanism (IRM) refers to the sensory mechanism selectively responsive to a specific external stimulus and responsible for triggering the stereotyped motor response. **Fixed action pattern (FAP**) is an inherent pattern of behaviour initiated by specific stimuli. It consists of species-specific, stereotyped movements e.g. following behaviour in goslings.

Object relations theory

According to object relations theory – the ego exists only in relation to other objects, which may be external or internal. 'Object' refers to both living persons and non-living concepts.

Melanie Klein was a major proponent of what came to be known as Object relation theory later. Other prominent theorists include Fairbairn, Kernberg, Guntrip, Winnicott and Balint.

Kleinian theory:

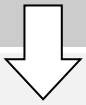
- Play interpretation was the major technique employed
- Maintained that oedipal development occurred earlier than what Freud envisaged
- According to Klein, an infant possessed instinctual knowledge of the body.
- Weaning is symbolically equivalent to castration
- Klein's stages are not age specific – but the PSP and DP are said to occur between 0-3 months (very early)
- Kleinian defenses – SIPDOG i.e. Splitting, introjection, projective identification, Denial, omnipotence and grandiosity

Winnicott's concepts:

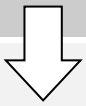
- Children's psychological development occurs in a zone between reality and fantasy called **transitional zone**. Play is an important aspect of development of a child.
- **Transitional object** refers to a soft toy, towel or any such objects that help in transition from ideal objects of fantasy to real objects which are not as reliable as those in fantasy. These serve as buffers against the loss, get invested with primary object's qualities e.g. mother's contact but remain under the control of the child.
- **Good enough mother** concept refers to the fact that a mother need not be perfect – but good enough to provide growth sustaining environment (**holding**).
- Parental control and impositions can lead to the development of a false self-different from the real self (**theory of multiple self-organizations**).

A flowchart describing Kleinian theory of infant's ('object-') relationship with the mother

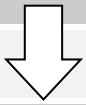
Soon after birth, fear of annihilation is present. This cannot be tolerated by the child and projects this destructive impulse to external objects.



Projection of both bad and good impulses occurs followed by splitting of the external world into good and bad. Cannot unify these elements into one. Bad objects include nongratifying bad breasts (parts). This leads to persecutory anxiety, and the child is said to be in Paranoid –schizoid position.



Later the child realizes that both good and bad things emanate from the unified single object (whole). At same time weaning occurs – perceived as a loss. Subsequent guilt develops for having destructive impulses against the mother. **Depressive** position – fear of loss of the love of object.



Reparation phase – creativity emanates as an attempt to repair damage done by 'destructive impulse'. Continues lifelong. In the absence of reparation, a maladaptive defense called manic defense can emerge characterized by denial of reality (refusal to take guilt), omnipotence and grandiosity.

3. Parenting practices

Parenting style is a psychological construct representing standard strategies that parents use in child rearing and includes the demands of children and response of parents. With respect to parenting, quality of care is more important than quantity of time spent. Parenting practices are specific behaviours.

Types of parenting: Maccoby and Martin described four parenting styles given in the table below. Corresponding to this classification, Baumrind described 3 response patterns in parents.

	Demanding	**Undemanding**
Responsive	*Authoritative*/Propagative	Indulgent (*Permissive*)
Unresponsive	*Authoritarian*/Totalitarian	Neglectful

- **Authoritative parenting**: parents are demanding but responsive; fits propagative parenting and concerted cultivation
- **Authoritarian parenting**: restrictive and punishment heavy; follow rules without explanation. High demand and low responsiveness.
- **Indulgent parenting**: Low demand, high response, the parents are permissible and lenient with few behavioural expectations placed on the child – parents try to be friends with a child. As adults, those with indulgent parents pay less attention to avoiding behaviours which cause aggression in others.
- **Neglectful parenting**: low demand, low response. These parents are detached and dismissive, not involved in the child's life. Also dismiss child's emotions and opinions.

	Authoritative	**Authoritarian**	**Permissive**	**Neglecting**
Control	Good degree of control	High control	Low or absent control	Low or absent control
Nurturance	Adequate warmth and nurturance	Poor nurturance	High degree of nurturance and leniency	Low nurturance; no warmth
Maturity demands	Age-appropriate demands expected	High degree of demands placed on growing child	No demands placed on growing child	No demands placed
Communication	Free flowing, positive	Poor flow of communication	Inadequate communication	Poor communication

Birth order is shown to have an effect on development through varying parenting practices.

- First-borns - get more parental time and have higher IQ, are more achievement driven and are more authoritarian, conservative and conformist
- Middle-borns - receive the least attention at home - have strong peer relationships
- Last-borns – receive most attention, get 'spoiled'; independent and rebellious

Effect of family dysfunction

Family structure has been known to impact the behaviour of children where those with single parents or large family sizes have been shown to increase behavioural issues. Single-sex couples and extended family involvement in upbringing do not cause the same issues.

Dysfunction in families cause discord, rejection of children due to processes such as disengagement and/or overprotection and over-involvement resulting in enmeshment.

According to the landmark **Cambridge study of Delinquent development** by Farrington et al, the most important childhood predictors (during age 8-10) of delinquency were antisocial child behaviour, impulsivity, low intelligence, low attainment, family criminality, poverty and poor parent child rearing behaviour.

Parental loss: Most children adapt well to parental divorce if financial support, reasonable contact with non-custodial parent and successful remarriage of single parent take place. If not, poor academic achievement, low self-esteem, 2-3 times more antisocial behaviour and higher rates of later life depression are seen.

- Children of all age groups are prone to short term behavioural difficulties after parental divorce – evident even in infants who may show changes in eating, sleeping and bowel patterns, with fearful or anxious responses.
- 3 – 6 age group often assume responsibility for parental separation
- 7 - 12 age group show decline in school performance
- Adolescents feel hurt, become angry and critical of their parents; they spend most time away from home as a reaction.
- Recovery usually takes 3 to 5 years.
- One third of all children have lasting psychological effects
- Boys are more affected than girls due to parental divorce; Among boys, physical aggression is a common sign of distress.
- Recent divorce or separation of the parents predicts suicide in children.
- 25% step families dissolve in 2 years, whereas 75% are harmonious.
- ADHD, Antisocial PD and conduct problems are more at homes without father.

- ✶ Children of divorced parents undergo divorce themselves twice more than children of un-divorced parent.
- ✶ Academic and social aptitude suffers due to divorce; asthma, injuries, headaches and speech defects are more common in divorced families.
- ✶ Divorce has more impact than death of a parent on psychological make up of a child
- ✶ Suicide rates for children of divorce are very high
- ✶ 25 % have adjustment problems at teenage.
- ✶ Parental death has somewhat lesser impact than parental conflict and separation. **Bereavement** causes increase in temper tantrums, depressive reaction (sadness, irritability), sleep disturbance. **Divorce** can cause all of the above but protective factors include positive temperament, relationship with other siblings & joint access.

Day care: Providing day care for more than 4 months at less than 1 year age for >20hrs a week can increase insecure attachment. If not, day care does not affect development adversely.

Adoption: Research shows that the earlier the age of adoption, the better is the outcome for the child. When a child is adopted during early childhood, then the chances of forming new attachments are better. Therefore early adoption is recommended as a matter of social policy. Children adopted before the age of 4 or 5 have been shown to do well generally. Although late adoption after the age of eight, does not necessarily lead to problems in adjustment, such children are more vulnerable and are at risk of developing future problems, like behavioural problems at home and school (Tizard and Hodges 1978).

Adopted children become aware of their adopted status most often between 2 to 4 years. Parental **disclosure** to children about their adoption reduces later psychological trauma.

Institutional care: Tizard and Hodges followed up a group of children who had been in institutions from infancy, adopted at age 4 and had been looked after by a number of carers who changed often. At age 8, most children had formed reasonably good attachment with their adoptive parents. At age 16, although the adolescents appeared to be functioning rather well, they showed a constellation of features termed as **ex-institutional syndrome**. These young people related better to adults than to their peers, were less likely to have a special friend, were less likely to be selective in choosing their friends and turned to peers less often for emotional support.

Intrafamilial abuse: Sexual abuse perpetrated by a parent can result in anxiety related symptoms, sexualized behaviour in the child, borderline personality disorder, substance misuse, dissociation and depression.

4. Temperament

This is an aspect of personality studied in infants. It describes individual differences in behavioural style. Certain aspects of temperaments remain stable over many years. Infant's negative emotionality (e.g. fear), and reactions to new situations (inhibition or neophobia) are the two most stable temperaments.

New York Longitudinal Study is a key study on childhood temperament conducted by Thomas & Chess. It is a thirty years (initially 6 years) longitudinal study of 138 children, observing childhood temperaments. It employed parental interviews to ascertain temperamental dimensions – 9 such dimensions have been used:

1. Activity
2. rhythmicity
3. approach/withdrawal
4. adaptability
5. intensity
6. threshold
7. mood
8. distractibility
9. attention span / persistence

Three behavioural styles were identified using the above 9 dimensions

1. ***Easy*** – rhythmic pattern of needs, adapts well, and active – 40%
2. ***Difficult*** – less predictable, uncomfortable with new experiences, negative mood, react intensely to stimuli, difficult to comfort – 10%
3. ***Slow to warm up*** children – adapts poorly to change, but less active and responds at low intensity – 15%
4. ***Ungrouped*** – 35%

Difficult temperament may offer some survival benefits as mother pays more attention – especially in tribal populations. Inhibition (approach/withdrawal dimension), according to **Keagan**, is a strongly inborn trait. **Behavioural inhibition** may be a precursor for later neurotic disorders including anxiety and depression. In an extreme form of inhibition called **neophobia**, a child appears frozen and withdrawn in novel situations.

Goodness of fit (Thomas & Chess) describes the reciprocal relationship between a baby's temperament and its social environment whereby a good match between the both results in positive development later. Chess and Thomas used the term especially to refer to the harmonious interaction between a mother and a child.

(Note that the concept of **Good-Enough Mothering** was proposed by Winnicott; according to him mothers provide a holding environment. A mother does not need to be perfect, but she must provide good-enough mothering.)

EAS model (Buss & Plomin 1984) describes three major dimensions: Emotionality – Activity – Sociability in children. EAS is a strongly biological model that views temperament as inherited personality traits exhibited in early life.

Resilience to mental illness

Resilience refers to the ability of children to function well in the face of adversity. There are 2 ways of considering resilience. 1. Resilience as a positive psychological outcome in the face of adversity. 2. Resilience as a dynamic process of psychological functioning that increases positive and reduces adverse outcomes in the face of adversity (Cummings, Davies, & Campbell, 2000).

Masten and Coatsworth (1998) reported that when children are faced with highly adverse situations (e.g. parental mental illness, family violence, poverty, natural disasters etc.), *personal characteristics* such as good intellectual functioning; appealing, sociable, easygoing disposition; self-efficacy, self-confidence, high self-esteem; talents; and faith can produce a positive psychological outcome despite adversity.

Certain *family characteristics* also convey resilience. These include having a close relationship with a caring parent figure, authoritative parenting (e.g., warmth, structure, high expectations), higher socioeconomic status and having extended family networks.

Extrafamilial factors that increase resilience include having bonds to supportive adults outside the family, being attached to prosocial organizations, and attending efficient schools/institutions.

5. Cognitive Development

Erikson's stages

Erikson proposed psychosocial developmental stages. These coincide with Freud's psychosexual stages but extend well beyond adolescence. It is not necessary that each stage must be resolved entirely before further progress. A mixture of positive and negative outcomes is noted for most people. However, if predominantly negative experiences accumulate at various stages of development, this may predispose to difficulties in life.

Crisis	Approximate Age
1. Basic trust vs. basic mistrust	Birth to 12-18 months
2. Autonomy vs. shame	18 months to 3 years
3. Initiative vs. guilt	3 to 6 years
4. Industry vs. inferiority	6 to 12 years
5. Identity vs. role confusion	Adolescence
6. Intimacy vs. isolation	Young adulthood
7. Generativity vs. stagnation	Middle adulthood
8. Ego integrity vs. despair	Late adulthood

In Erikson's model autonomy refers to children gaining more control over activities and acquiring new skills. This is crucial for building self-esteem, failing which a sense of **shame** is felt.

During 6 to 12 years of age we are capable of learning, creating and accomplishing numerous new skills and knowledge, thus developing a sense of **industry**. This is also a very social stage of development, and if we experience unresolved feelings of inadequacy and inferiority among our peers, we can have serious problems in terms of competence and self-esteem.

Identity vs. role confusion. This stage occurs during adolescence between the ages of approximately 12 to 18. Up to this stage, according to Erikson, development mostly depends upon what is done to us. But from teenage onwards, our development depends primarily on what we do. Teens need to develop a sense of self and personal identity. During adolescence, children explore their independence and start to form a sense of self. This phase of transition from dependent child to an independent adult is associated with confusion and insecurity. Teens also experiment with different social roles at this stage. According to Erikson, this is important to the process of forming a strong identity and developing a sense of direction in life.

The inherent strength of young adulthood is love, and the major task is **intimacy** and formation of a future bond partner.

A middle aged adult seeks satisfaction through productivity in career and family / social network. This is referred to as **generativity**.

An older adult reviews/cherishes life accomplishments prepares for end of life by pursuing lifelong interests etc. This is referred to as **integrity**

Piaget's model of cognitive development

According to Piaget, a **schema** is defined as the basic building block or unit of intelligent behaviour. Schemas consist of organized past experiences to understand future experiences. We have physical schema such as bike riding schema and mental schemas such as addition, multiplication and division schema.

Within each developmental stage, functioning is generally internally consistent and stable and thus said to be in **equilibrium**. Stage-to-stage transformation occurs as a result of interaction with environment, whereas existing schemas cannot solve the environmental realities.

Adaptation is the process of fitting schemas to environmental information. Adaptation can occur either as assimilation or accommodation. In **assimilation** new information is incorporated into existing schemas without restructuring the schemas. In **accommodation**, the schemas are restructured to 'accommodate' newly learnt information.

Equilibration is achieved when all information properly fit into the schemas via either processes of adaptation. During each developmental stage, the child will experience cognitive disequilibrium, which through adaptation, gets solved, and equilibration results. Each time that equilibration occurs, the child produces more effective schemata or mental structures.

Approx. AGE (not delineated by Piaget)	STAGES	FEATURES
0 to 2 years	**Sensorimotor**	Exercising reflexes (0 to 1 m) to smoothen them; Primary circular (1 to 4 m) reactions where reflexes extend to objects; secondary circular (4 to 10 m) where goal direction seen; object permanence starts by 9m; coordinated actions with added element of curiosity forms tertiary circular (12 to 18 m) reactions – here novelty is sought. Mental combinations occur; thoughts dominate actions.
2 to 7 years	**Pre-operational**	Preconceptual stage 2 to 4 years; intuitive stage 4 to 7 years.
7 to 11 years	**Concrete operational**	Ability to decentre, conserve, seriate and declining egocentrism noted. Perspective taking starts to develop. But transitivity tasks still pose a challenge. E.g. '4>2, 2>1, which is the greatest of all?' - is still difficult.

> 11 years	**Formal operational**	Here manipulation of ideas and propositions are seen – 1st order operations; soon, reasoning solely based on verbal argument construction develops – 2nd order operations. Hypothetico-deductive reasoning develops in a proportion of children after age 12.

Sensorimotor stage: (SPIRO)

- **S**ymbolic thought: Language starts developing and thought starts to dominate actions.
- **R**epresentational **P**lay: Mimics one object with another e.g. cup for a hat.
- **D**eferred **I**mitation: remembers an act and replays it later.
- **R**ecognition of self: Primitive self recognition begins.
- **O**bject permanence: Understanding that object that disappears from field of perception has not ceased to exist; if searched well this object can be found or it will reappear. Hence peek-a-boo games are understood and enjoyed. Initially this is limited as the hidden objects are searched at where they were last seen (around 9 to 12 months); not at where they were hidden. Around 18 months invisible displacements are inferred and object permanence is completed.

Preoperational stage: (FAT PILES)

- **F**unctional attribution: Objects are referred to by their function rather than appearance.
- **A**rtificialism: 'Sky is blue because someone painted it'
  - & **Animism**: Inanimate objects are treated as living objects.
- **T**ransductive reasoning: Cats have 4 legs, Dogs have 4 legs. So cats and dogs are the same (called Von Domarus law).
  - & **Telegraphic speech**: No functional propositions noted but verbs and nouns are used
- **P**henomenalistic causality: In a similar logic to transductive reasoning, causality is inferred if two events occur with some temporal association e.g. lightning and rain come together; hence lightning brings rain.
- **I**mminent justice: See moral development
- **L**ack of **seriation**, **conservation**, and **reversibility**:
  - **Seriation** is the ability to sort or categorise based on dimensional variations of items. In **centration** only single dimension can be focussed at one time (akin to syncretic thought, see below). **Conservation** refers to the ability to perceive that a quantity (such as count, weight, volume etc) is unchanged if the same amount of a material is

transformed into a different shape or structure. E.g. 1 litre of water remains the same 1 liter irrespective of whether it is present in a half full 2 litre bottle or two complete 500ml bottles. Concept of **compensation** refers to the fact that magnification in one dimension and reduction in another dimension can nullify each other's effect. **Reversibility** refers to the ability of mentally calculating and understanding that what is done can be undone without loss of material.

- **Egocentrism**: This does NOT refer to self-centredness or selfish attitude. It refers to the restricted ability of viewing the world from a single point of view at this developmental stage. This was demonstrated using the Mountains task where a child at this age group could not say what a person would see from other side of the desk when only one side of a toy mountain was visible from each view.
- **Semiotic function**: Signifiers are symbols and signs that represent or stand for something else. For example, drawing a matchstick man. Thus signifiers represent a meaning, serving semiotic function. This is vital for developing play activities.
- **Syncretic thought**: Links neighbouring objects and events on the basis of common instances e.g. red square with red sphere with blue sphere with blue cube etc.

Concrete operational stage:

Conservation of **liquid** develops around 6 years, followed by conservation of length, count, weight and volume (around 11 to 12 years) in a **vertical decalage** fashion (i.e. not parallel but one by one development of these abilities). Note that if the same question about quantity is asked before and after manipulation of materials in front of a child, the child answers differently for the second question as he/she interprets that the question is asked twice as the answer given was wrong in the first instance. If the pre-transformation question is dropped, the conservational ability could be demonstrated at earlier stages. Also using an external 'accident' such as **naughty teddy** that disturbs a heap of coins, conservation could be demonstrated earlier than what Piaget thought.

6. Language development

Basic speech sounds are called phonemes. In English language, there are 46 speech sounds. Most children can differentiate speech sounds before being able to produce them. The basic meaningful part of the language is called morpheme. The rules for combining words into phrases and sentences are called syntax.

Language is slower to develop in boys, in twins, in large families, in those from social classes 4 and 5 and those that lack speech stimulation e.g. deaf and neglected children.

Stages of language development:

The pre-linguistic state (0 to 12 m): Crying is an important form of communication. A one-month-old child is able to distinguish speech sounds although these phonemes are almost identical sounds. This categorical speech perception is supposed to be innate. By six weeks, the child starts cooing. By six months, **babbling** is seen. Babbling is nothing but the repetitive production of speech sounds. Spontaneous babbling refers to the situation when the child enjoys making these sounds alone. All babies around same age irrespective of the culture start bubbling. Even the deaf babies of the deaf-mute parents start marbling but stop at 9 to 10 months. The phonemic expansion refers to the expansion of production of phonemes even if such phonemes are not seen in the native language.

One word stage (12 m to 18 m): Jargon words and babbling continue up to 18 months. First words are often self-invented but carry meaning and consistently match with the same meaning. There is a clear intention to communicate. Earliest words are **context bound**; sometimes they do not have any communicative purpose but are used as performatives to refer to actions. A child says 'teddy' only when the teddy is thrown up into the air while playing; thus teddy refers to 'throw up' action rather than the doll. **Holophrases** are one-word substitutes for whole phrases or sentences. At this stage, a child understands more words than it could produce. Gradually words get decontextualised and fall into one of the following functions;

1. Nominals – specific e.g. Sarah, or general – e.g. ball pen
2. Action words e.g. bye, look.
3. Function words e.g. the for, what, etc. propositions and grammatical functions.
4. Modifiers – e.g. red, big, etc.
5. Personal and social function – e.g. oops, ouch, etc.

Two-word sentences / stage 1 grammar (18 to 30m): Telegraphic speech is seen where meaningful words are used without connecting words. At this stage adults interact with children

in a '**motherese'** – short simple raised pitch paraphrased language directed at infants. As object permanence is achieved by this stage, words start to have representational functions.

Stage 2 grammar (> 30m): Mean length of utterances increase largely due to the use of function words – propositions, etc.

Noam Chomsky: Children are born with an **innate language acquisition device**. **Transformational grammar** is important in understanding language development. All languages have a surface structure where the syntax is accurate and actual words are used to construct language; and a deep structure where the most semantic sense is made without similar syntactical rules. A single surface structure can have several deep structures. Children are born equipped with the ability to decipher the transformational grammar of deep to surface structure conversion. Hence, years up to puberty are sensitive though not critical fro language development.

Social interaction view of language development: Adults such as mother act as LASS (**language acquisition support system**). This is essential for the function of language is social interaction.

AGE & LANGUAGE DEVELOPMENT

3 months-babbling

9 months-repetitive babbling

12 months-speaks three words

18 months-speaks up to 40 words.

24 months-telegraphic speech, grammatically pairs words and vocabulary more than 240 words

36 months-early comprehension of grammar and syntax

48 months-correct use of grammar

60 months-language akin to adult speech

The **elaborate language code** is characterized by longer, complex sentences that are context-independent. It focuses on the past and future, employs pronoun 'I' commonly and allows for the expression of abstract thought. **Restricted language code** is characterised by short, incomplete sentences, which tend to be context-dependent, frequently uses like 'you know', focuses on the present, employs pronoun I rarely and has little room for expressing abstract thinking. People in lower socioeconomic classes more commonly use the restricted code whereas the middle class and upper class children often use elaborate language code. Such differences emerge from the influence of social interactions in language development.

7. Social competence and peer relationships

Through friendship, children learn cooperation, sharing and conflict management. They also learn empathy and group belonging. Studies suggest that peer rejection can later result in depression, school drop out and other psychiatric issues. (Cairns, Cairns and Neckerman, 1989)

Social competence is a complex concept involving social, emotional, cognitive and behavioural skills. It is the foundation upon which ability to interact with others is built and also perceptions of own behaviour is developed. There are several approaches involved.

Peer regard/status	Define social competence based on popularity amongst peers
Social skills	Behaviours demonstrating social skills are used to determine social competence
Relationship	Social competence is based on the ability to form and quality of relationships
Functional	Context-specific, it is concerned with the identification of social tasks

Peer groups are comprised of children of similar age, background, social status and often with similar interests. This primary social group can influence several behaviours and beliefs. Children look to join peer groups that accept them, even if they are involved in negative activities disapproved by parents.

Peer acceptance: the extent to which a child I viewed by peers as worthy and likeable companion. This is assessed using sociometric techniques

Popularity: Popularity is not the same as having many friends - many popular children do not count having a large number of friends.

1. **Popular** children: liked by many disliked by few
2. **Rejected** children: disliked by many liked by few
3. **Neglected** children: neither liked nor disliked (few nominations in sociometric measurements in a classroom)
4. **Controversial** children: liked by many but also disliked by many
5. **Average-status** of acceptance: nearly 1/3rd of the class – liked or disliked by a moderate number of peers

Popular children show the following characteristics:

- ✶ Respond positively in unfamiliar environments

- ✶ Initiate interaction in new social groups
- ✶ Comment constructively in groups
- ✶ Blend smoothly with new peers
- ✶ Pleasant temperament
- ✶ Academic skills
- ✶ Display ease when interacting with opposite gender

8. Moral development

Freudian theory: According to Freud, boys have unconscious wishes to compete with the father for mother's love. This leads to **castration anxiety** as fear of being punished for competing sets in. This anxiety drives the repression of such desire and leads to identification with one's father from whom the superego morality is incorporated; thus moral development is achieved via the development of the superego. In girls Oedipus complex is not seen; instead **penis envy (Electra complex)** is noted. Father is the love object here, and unconscious wish for having a baby from the father is present, but without a strong anxiety as seen in boys, the identification with mother and imbibing of superego occurs. Hence, Freud claimed that superego or morality is weaker in women than men.

Piaget described qualitative differences in older vs. younger children in terms of morality that was based on the ability of older children to have **social perspective** (an extension of the Theory of Mind concept); he did not describe stage-by-stage development of morality. Cognitive development is essential but not sufficient for moral development. Moral development lags 2 years behind the cognitive development.

Piaget's Moral Development Theory	
5 to 9 years	Older than 10 years
Unilateral respect for the external law: seniors make rules; they are sacred and should not be broken, but get violated periodically for pleasure.	**Mutual respect** for the self-invented law: could be changed by consensus and for fairness.
External responsibility holds for crime; severity of outcome or loss decides the degree of punishment warranted.	**Internal responsibility** holds for crime; intent or motivation decides the degree of punishment warranted.
Moral realism: Strong penalty should be paid for any crime; can accept collective punishment to deliver justice (punishing the wrongdoer is more important than not hurting the innocent)	**Moral relativism**: Punishment should match the crime; does not accept collective punishment.
Imminent justice: Wold is just – a misfortune will punish the deserved for a misdeed.	No imminent justice.
Heteronomous morality: Subject to rules written by others.	**Autonomous morality**: Rules can be self-made.

Kohlberg's theory of moral development: This is a stagewise process where reasons for making a judgment in a hypothetical experiment (**Heinz Dilemma**) are studied in children; reasons are more important than the actual judgment made. On this basis, Kohlberg identified 3 levels and 6 stages.

- **Level 1 Pre-conventional morality (7-12 years to middle childhood)**: In this stage, the children decide right or wrong according to the consequences. If an action leads to punishment it must be bad and if it leads to reward it must be good.

  i. Punishment and **obedience** orientation: Obedience to rules to avoid punishment
  ii. **Reward** orientation/ Instrumental relativism: What brings rewards is right. 'tit for tat' approach seen.

- **Level 2 Conventional morality (approximately 13-16 years)**: Here the children believe that social rules and the expectation of the others determine what acceptable or unacceptable behaviour is.

  iii. **Concordance** orientation: What pleases others is right. What the majority thinks right is right. Also called Good boy/good girl orientation. Conforms to avoid disapproval and meet expectations of others. Being good is important and having good motives and showing concern
  iv. Social order or **Authority** orientation: Upholds laws and social rules to avoid the censure of the authorities and feelings of guilt about not doing one's duty. Maintaining social order is the goal.

- **Level 3 Postconventional morality (approximately 16-20 years)**- Here what is right is based on an individual's understanding of universal ethical principles. These are often abstract and ill-defined, but it might include the preservation of life at all costs and the importance of human dignity

  v. Social contract or **legalistic** orientation: Actions guided by principles commonly agreed by one's group on as essential to public welfare (relative values) and democracy is upheld while individual's life is given more respect than written codes of law.
  vi. **Universal Ethical** orientation: Actions guided by self-chosen ethical principles. Laws and social principles usually valid because they are based on these principles. Social rules can be broken if universal morality is not upheld.

Level 3 cannot be considered a part of the normal or expected course of development and instead represents a philosophical ideal. Only 15% eventually achieve level 3. Formal operational thought is necessary but not sufficient to achieve level 3 morality. Kohlberg's stages are criticized to be

androcentric (all male sample) and Eurocentric. Though they are well correlated with 'moral reasoning', they are not so well associated with actual behaviour.

Eisenberg's stages: Both Kohlberg's and Piaget's theories were based on the prohibition of the wrong; Eisenberg's was based on **prosocial** reasoning where helping or altruistic behaviour was studied.

Social learning theory argues that while actual reinforcement is not needed for 'learning' about morality, the performance of a moral deed can be reinforced (either directly or vicariously). Vicarious punishment is more effective than vicarious positive reinforcement in this regard.

9. Emotional literacy and fears

Emotional literacy is a term used interchangeably with emotional intelligence (Steiner, 1997). Components of emotional literacy includes:

1. Knowing your feelings
2. Having a sense of empathy
3. Learning to manage our emotions
4. Repairing emotional problems
5. Emotional interactivity – putting it together

Emotion regulation describes an individual's ability to gauge the appropriate level of emotional response required and respond to environmental stimuli with a range of emotions in a controlled manner (Panfile and Laible 2012).

Emotion regulation develops throughout the lifespan (Cole et al., 2009). Infants have limited emotional regulation, shown by **gaze aversion and vocalising**. Around 1 year of age children are able to unconsciously regulate their emotions. Between 3 and 5 years (at kindergarten) children may tolerate ordinary, brief frustrations and handle minor disappointments (Cole, 1986; Cole et al., 2003).

In a detailed longitudinal study, (**Development of Toddlers Study** (D.O.T.S.)) Cole et al. observed the child in spontaneous situations at 4 time points (18, 30, 36, 42 months). Their observations are tabulated below:

18 to 24 months

- were quick to express anger
- slow to distract themselves
- might try to distract themselves but only briefly
- bid to mother but angrily

36 months

- quickly bid to mother (but thru *words* not anger)
- were somewhat quicker to distract
- anger was briefer, distractions longer

48 months

- quickly & briefly bid to mother (verbally)
- quickly distracted themselves
- eventually focused on gift & then showed anger

Imaginary companions are often a part of emotional development.

- ✶ Preschool years mostly (between 3 to 10 yrs age, 50% may have),
- ✶ Usually in children with above-average intelligence
- ✶ Usually in the form of persons rather than animals/things
- ✶ Usually friendly and help to reduce loneliness and anxiety.
- ✶ Mostly disappear by age 12, occasionally persist into adulthood.

Developmental fears

- ✶ Fear of animals-age 3
- ✶ Fear of the dark-age 4 or 5
- ✶ Fear of imaginary creatures - 5 plus
- ✶ Fear of open spaces arises in later childhood or adult life
- ✶ Fears that arise in late childhood or adult life: fear of sex/open spaces.
- ✶ Teenage onwards-failure, illness and death
- ✶ Fears that show no particular age trend-fear of snakes or storms

Simple fears are often linked to early negative childhood experiences or learnt from other family members (such as a sibling's fear of spiders may influence a child). Maintenance of phobias is due to avoidance of the anxiety-provoking stimuli relieving unpleasant emotions, which becomes a reward itself.

Children of age 3 to 6 yrs are aware of their body and show a preoccupation with illness or injury, - every injury must be examined and cared for – hence this phase is also called **Band Aid Phase.**

10. Sexual development

Gender identity typically forms around the age 3-4 and remains established. Gender typing describes the process where an individual acquires a sense of gender-related traits within the society they are born. It usually starts with clothing at a young age.

Gender role: behaviour an individual engages in that identifies with their gender e.g. use of cosmetics.

Theories of Gender Identity development

- According to **Social Learning Theory** of gender development, girls and boys learn to behave differently because the society treats them differently (pink dresses for girls, Barbie dolls for girls, etc.). Fathers treat children in a more gendered way than mothers. This sex typing is reinforced by further observational learning by the child and reinforcement that results on accomplishing gender specific developmental tasks.
- **Cognitive Developmental Theory** stresses the importance of child's participation in gender identity. A child's discovery about his/her biological sex leads to identification with a group and conformity as a result. Thus three stages have been identified:

    1. Basic gender identity / **gender labelling**: Around age three, a child understands she/he is female or male.
    2. **Gender stability**: By age 4 – 5 recognise that gender is retained life-long and will not change!
    3. **Gender constancy** / consistency: Age 6 -7 understand that gender is immutable even if physical changes are carried out. It is a type of *conservation* achieved akin to Piaget's cognitive development

- According to **Gender Schema Processing Theory**, gender identity alone provides children the motivation to assume sex-typed behaviour. Following this they observe and learn to be of a specific gender in the society. Thus, a gender schema of the particular culture gets deeply incorporated and serves as a standard for comparison.

Sex drive exists from birth but increases in adolescence due to raised androgen secretion. **Sexual orientation** is explored during adolescence. There are arguments for and against the biological determination of sexual orientation vs. shaping in childhood.

Sexual Behaviours in Childhood	
Preschool (age <4 years)	✶ Exploring private parts through touch and rubbing or showing to others ✶ Trying to touch women's breasts (including mother) ✶ Exposing oneself and attempting to see other exposed people (adults and children)
Young Children (approximately 4-6 years)	✶ Stimulating genitals (masturbating) sometimes when others are present ✶ Kissing, or holding hands ✶ Talking about genitals without grasping the meaning in full ✶ Exploring private parts with peers e.g. "playing doctor", "I'll show you mine if you show me yours."
School-Aged Children (approximately 7-12 years)	✶ Masturbating privately ✶ Playing courtship games ("mummy and daddy") ✶ Gazing at pictures of naked people; viewing sexual content in media (✶ Wanting more privacy ✶ Showing reluctance to discuss sexual issues with adults ✶ Being sexually attracted to peers

11. Adaptations in adolescence & adult life

Adolescence is the period between childhood and adulthood and is described in many of the developmental models (Piaget – formal operational, Freud - genital stage, Erikson – identity vs. role confusion).

Conflict with parents is common during this time due to developing a sense of self and autonomy. This could be attributed to a **second separation-individuation** phase.

Anna Freud described **affective instability** as the oscillation between behavioural and affective excess and scarcity during adolescence induced by endocrine changes, sexual maturity and instability of ego defenses.

Erikson described **adolescent turmoil** as a temporary maladaptive state that was due to identity diffusion. According to him all adolescents passed through this state. But later studies (Offer & Offer 1975) showed that while upheaval and turmoil are common in adolescence, they might not occur in all adolescents. Nearly 23% showed continuous linear development during adolescence, while 35% were late bloomers who were less introspective and had some frictions with their families. Around 21% had recurrent conflicts with their parents and chose less competitive careers.

Marcia's theory on adolescence: A mature self-identity is possible only if an individual experiences several crises, finally arriving at a stage of commitment. Successful maturation during adolescence depends on both the degree of crises faced and commitment achieved, with different levels of maturation as shown in the table below.

		Degree of crises	
		HIGH	**LOW**
Degree of commitment	**HIGH**	Identity achievement	Foreclosure
	LOW	Moratorium	Role confusion

- **Identity achievement**: Most mature achievement – most desirable.
- **Foreclosure**: Avoids anxieties by prematurely committing to safe and conventional parental and societal goals and beliefs.
- **Moratorium**: Experiences height of crises but postpones decisions until alternative identities are tried.
- **Role confusion**: a unresolved state of adolescence

Puberty trends:

- ✶ In the UK the average age of onset of puberty in males is 11.2 years; for females it is 11 years. Menarche on average is at 12.5 years for females.
- ✶ There is a general trend for falling in menarcheal age globally over last 50 years. Compared to US and other European countries this falling trend is smaller in UK (by about 6 months in 30 years).
- ✶ Social stress is also a puberty accelerator, with familial disruption and father absenteeism, being one of the most effective stressors.
- ✶ Precocious puberty is suspected in boys before age 9, and girls before age 8.

As children develop through adolescence into adulthood, they hold increasingly complex orientations to the self and to the interpersonal world. Jane Loevinger described 9 stages of ego and personal identity development involving childhood, adolescence and adulthood.

1. Presocial – the baby is unable to differentiate itself from the world
2. Impulsive – the child is concerned with bodily impulses
3. Self-protective – the child has a notion of blame but externalises to the situation or other people
4. Conformist – around school age, they conform to socially approved codes/the norm
5. Self-Aware – (conscientious-conformist) transitional stage – increased self-awareness and self-criticism; deepened interest in interpersonal relations
6. Conscientious – internalisation of rules is complete; goals/ideals are acknowledged, and there is a new responsibility – feel guilt for hurting others rather than rule-breaking
7. Individualistic – respect for individuality and interpersonal ties
8. Autonomous – 'synthesizers' – able to conceptually integrate ideas
9. Integrated – rarely attained stage "learning is understood as unavoidable…the unattainable is renounced."

Pairing occurs often within the same cultural and socioeconomic background (**homogamous mate selection**). **Equity theory** suggests that individuals consider the cost-benefit ratio for each person in a relationship; **reinforcement theory** suggests that individuals chose their partners on the basis of reinforcement of attraction with rewards.

The midlife transition occurs around age 40 to 45. The term '**Midlife Crisis'** was coined by Elliot Jacques who distinguished it as a critical phase in development as the transition between the forties and early sixties, and is often associated with coming to realise mortality, unrealized goals, menopause or children leaving home. **Downshifting** refers to voluntary opting out of a

pressurized career and giving up well-paid job for more fulfilling life (anti-urbanism). **Empty nest distress** refers to the feeling of loneliness when children leave home.

Bereavement is usually used to describe loss if a person, but can be loss of anything e.g. marriage, employment, and refers to being in a state of mourning (a process influenced by culture and society in which grief is resolved).

The classic work on stages of grief came from Erich **Lindemann,** who studied 101 bereaved people and published in 1944; an article titled "Symptomology and Management of Acute Grief". In this article he described a set pattern of reaction to a loss event (grief): After an unexpected death, there is the initial **shock** that lasts 10-14 days. After the initial shock comes a period of intense **sadness**, and the grieving person may withdraw from social contact. Next comes **anger**, as the grieving person seems to '**protest'**; the unexpected death. Finally, within a year or so, the grief is **resolved,** and the person returns to normal. These stages were further refined by Parkes.

5 stages of bereavement (Parkes)

1. Alarm
2. Numbness
3. Pining for the deceased (illusions or hallucinations of the deceased can occur)
4. Depression
5. Reorganisation (recovery)

Physiological events such as **pregnancy** and childbirth could also be stressful in adult life. The psychological stress during pregnancy can have physiological implications on the growing fetus. Release of corticotrophin releasing hormone (CRH) from the placenta increases with stress, and with this an increased risk of intrauterine infection, preterm labour and low birth weight is seen. Pre-term infants are susceptible to complications later such as developmental delay and increased rate of mortality. Babies subject to stress in utero can have the difficult temperament and are irritable. There is additional data to suggest that stress in utero can result in increased risk of chronic health issues in adulthood e.g. hypertension and diabetes.

Psychological symptoms following **childbirth** include intrusive thoughts, avoidance, anxiety, depression, social dysfunction and somatisation. Significant psychological distress is noted in 37% of mothers and 13% of fathers. Factors predicting acute maternal psychological distress included being a single parent, multiparity and previous traumatic birth. At six week and six months, psychological distress symptoms fell to that of general population level indicating that in most people, this is a short-term distress.

12. Adaptation with ageing

Age is the first social category learnt by a child even before number concept develops.

The effect of ageing on cognitive abilities: The mass of the brain reduces with ageing (in some cases this can be pathological). Memory is often cited as the cognitive function most susceptible to decline with age – particularly working memory and incidental memory. Attention also declines with age.

The effect of ageing on body physiology: Bone loss results in a reduction of mechanical strength, collagen fibres deteriorate causing loss of elasticity of the skin, and the efficacy of organs deteriorates with age.

Ageing and socioeconomic independence: Older adults are an economically vulnerable group and with an increasing older population, there are concerns they may outlive their financial resources. The decline in socio-economic status can result in a decline in their physical and mental health. Reduced mobility from physical health problems impacts their independence and ability to maintain social ties.

Old age – theories of role change:

1. Social **disengagement** theory: Mutual withdrawal of society and the individual occurs; increased individuality and shrinking life space are inevitable moves towards death.
2. Social (*non*) **reengagement** theory (aka activity loss theory): Ageist society reduces the social interaction that older adults can have; withdrawal is not mutual but forced.
3. Social **exchange** theory: Age robs people of the ability to engage in reciprocal roles; retirement is a special social contract wherein productivity is exchanged for increased leisure and decreased responsibilities.
4. Socio-emotional **selectivity** theory: Wise investment of social energy in old age is to limit social interaction to those who are most familiar.

Phases of retirement:

1. Pre-retirement phase: Increasing anxiety with the retirement of friends and colleagues.
2. Honeymoon phase: Immediate post retirement phase where increased freedom is enjoyed.
3. Disenchantment: Slowing down occurs, feels let down, worse if inadequately prepared for retirement.
4. Reorientation – explores new avenues, more realistic.
5. Stability –makes choices, mastery attained in chosen leisure.
6. Termination – frailty, death.

13. Genetic influences on development.

Brain development is genetically programmed. For example, the cortical thickness in lateral prefrontal, medial prefrontal, parietal, and temporal cortices is highly heritable. Regions crucial for evolutionarily advanced functions (such as higher-order cognition, sociality, and language) show both high heritability and an increase in heritability with age during development. This pattern is similar to the age-related increase in heritability for traits such as IQ (discussed earlier). This could be either due to an age-dependent gene expression or due to gene-environment interaction.

Gene-environment interaction refers to the influence of the relationship between genotype and the environment in shaping a phenotype. Consider the example of **phenylketonuria**, a genetic condition caused by the lack of an enzyme that breaks down the amino acid phenylalanine, resulting in intellectual disabilities. However, the usual dietary environment of a child can be altered by the use of phenylalanine-free foods, which results in normal development.

Waddington proposed the concept of **canalization** in developmental genetics. Certain behaviour traits are strongly genetically determined (canalized), so development follows these behaviours (e.g. crawling). Other behaviours are poorly canalized (e.g. cycling), so environmental factors will influence these traits. Further, some traits appear to be heavily canalized early in development (e.g., language development), but less so later on (e.g., reading ability).

Gottesman put forward the notion of a **range of reactions**. The genetic make-up of a child does not shape any behaviour in its entirety; instead genes only set limits (or range) within which the individual variability is shaped by the environment.

Scarr & McCartney proposed the concept of '**niche-picking'** in developmental genetics (1983). The genetic make-up of a child does not contribute to skills or behaviours as such; instead genes only contribute to propensities toward certain skills and abilities; children then seek activities that are compatible with their genetic endowment

Three types of G-E interaction are notable:

- ***Passive GE-Interaction:*** A child's environment is influenced in part by parental genes which are in turn correlated with the child's genes. Thus, the phenotype is not a direct result of the child's genotype; but results from environmental influences that are indirectly associated with the parental genes. This kind of passive G-E will show decreasing influence over development. e.g. a boy who does not inherit the same tallness and

physique as his father may still develop into a good basketball player in line with the father due to repeated exposure to basketball sessions.

- ***Evocative GE-Interaction:*** A child's environment is influenced in part by genetically shaped behaviour. Thus, the phenotype is not a direct result of the child's genotype; but results from environmental influences evoked by the child's own genes. This kind of evocative G-E will show stable influence over development. e.g. a girl who inherits impulsivity may evoke an abusive parental reaction, leading to later depression, though depression itself is not inherited.
- ***Active GE-Interaction:*** A child's environment is influenced in part by an active choice of the child to complement genetically shaped interests. Thus, the phenotype is not a direct result of the child's genotype; but results from environmental influences that are actively associated with the child's own genes. This kind of active G-E will show increasing influence over development. e.g. a girl who inherits impulsivity may choose to gamble, losses in which may lead to later depression though depression itself is not inherited.

14. Neuroimaging and neurodevelopment

(This section is best read in conjunction with the section on neuroimaging in the Clinical Examination chapter)

A neural tube is seen 2-3 weeks after the formation of the human fetus. By week 5, the ectodermal tissues differentiate to precursors of different brain regions. This is followed by birth neurons from stem cells (at ventricular proliferative zone, by week 8), neuronal migration (week 12-20), formation and pruning of axons, dendrites and synaptic contacts, myelination of axons and apoptotic removal of excess cells.

One of the earliest neurodevelopmental events that can be visualized using neuroimaging in human fetuses is the **migration of neurons**. Around 17 weeks' of gestation, a transient layer of cortical **subplate** of migrating neurons is visible beneath the cortex; by 20 weeks the subplate withers away and replaced by more permanent cortical sheet at 24 to 28 weeks.

A cortical folding pattern consisting of sulci and gyri become visible on fetal MRI by 20 weeks. Around the 28th week of gestation, the neuronal count in the human brain is at its peak - around 40% greater than in the adult. Dendritic formation accelerates at this time (but cannot be seen in MRI), and along with the disappearance of the proliferative zone and cortical subplate, an increase in cortical thickness is notable on fetal MRI.

Synaptogenesis peaks around 34th week of gestation in (~ 40,000 new synapses formed per second) and continues in postnatal life alongside active synapse elimination; eventually the net number of synapses begins to decrease at puberty (pruning). Synaptic pruning cannot be observed directly with neuroimaging, but a prominent progressive cortical thinning of frontal and parietal cortices is noted in MRI studies during adolescence and is attributed to synaptic pruning. An indirect indicator of synaptic removal is the glucose metabolism measured using positron emission tomography (PET).

Myelination of the visual cortical white matter begins prenatally (last trimester); by 9 months of postnatal life, myelination extends to frontal cortex (posterior-to-anterior maturation starting with sensory pathways motor pathways, and finally the higher-order association areas).

Landmark longitudinal studies of brain volumes in children were conducted in the Child Psychiatry Branch of the NIMH, USA. These studies report the following observations

1. **White matter volume** increases linearly up to age 20 years in all brain regions;
2. Frontal, parietal, and temporal **gray matter volumes** follow an "inverted U-shaped" developmental curve (increase before adolescence, frontoparietal peak at 12 years and temporal peak at 16 years of age, followed by universal reduction thereafter)

3. **The cortical thickness** decreases with advancing age in "back-to-front" progression starting from sensorimotor areas, progressing to the dorsal parietal, superior temporal, and dorsolateral prefrontal cortices later. In other words, sensorimotor area mature earlier than higher-order regions.

Cortical thinning may occur either due to synaptic pruning or myelination (with myelination, the brain tissue that is identified as white matter in MRI scans increase in proportion, reducing the relative grey matter thickness).

Diffusion Tensor Imaging, a technique used to study the integrity of white matter tracts, reveals that in children, with advancing age, the directionality of diffusion in white matter pathways continues to increase especially in the prefrontal regions and in basal ganglia. This suggests that frontostriatal systems myelinate progressively during adolescence. DTI studies also show that the frontotemporal pathways may continue to myelinate until age 30 years.

Magnetic Resonance Spectroscopic measure of N-Acetyl-Aspartate, an indicator of neuronal integrity, reveals low levels around birth that increase rapidly during the first 2 years of life, slowing down thereafter; this may represent synaptogenesis during childhood.

Functional MRI studies reveal age-related increases in activation left frontal and temporal cortices (language areas) supporting the expansion of reading and phonological skills during childhood.

DISCLAIMER: This material is developed from various revision notes assembled while preparing for MRCPsych exams. The content is periodically updated with excerpts from various published sources including peer-reviewed journals, websites, patient information leaflets and books. These sources are cited and acknowledged wherever possible; due to the structure of this material, acknowledgements have not been possible for every passage/fact that is common knowledge in psychiatry. We do not check the accuracy of drug-related information using external sources; no part of these notes should be used as prescribing information.

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